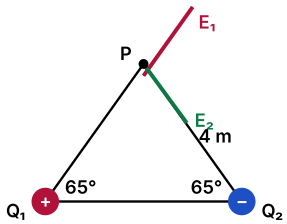
2do parcial · I · Campo en triángulo

E_R y θ en P

$+Q_1 = 7\times10^{-9}$ C (abajo izq.), $-Q_2 = 4\times10^{-9}$ C (abajo der.), lados = 400 cm ÷ 100 = **4 m**, ángulos de base 65°.
E_1 **sale** de $+Q_1$ → 65° · E_2 **entra** a $-Q_2$ → 360 − 65 = 295°



✍ A MANO	📱 CALCULADORA
$E_1 = 9\times10^9\cdot7\times10^{-9} / 4^2$ $10^{9-9} = 1$	9 × 7 →63 · 4 x² →16 63 ÷ 16 = →3.9375
$E_2 = 9\times10^9\cdot4\times10^{-9} / 4^2$	9 × 4 →36 · 36 ÷ 16 = →2.25
E_1 = 3.94 N/C · E_2 = 2.25 N/C	
$E_1x = 3.9375\cdot\cos65 = 1.664$ $E_2x = 2.25\cdot\cos295 = 0.951$	65 cos × 3.9375 = 295 cos × 2.25 =
$E_1y = 3.9375\cdot\text{sen}65 = 3.569$ $E_2y = 2.25\cdot\text{sen}295 = -2.039$	65 sin × 3.9375 = 295 sin × 2.25 =
$\Sigma Ex = 1.664 + 0.951 = 2.615$ $\Sigma Ey = 3.569 - 2.039 = 1.530$	1.664 + 0.951 = 3.569 − 2.039 =
$E_R = \sqrt{(2.615^2 + 1.530^2)}$	2.615 x² + 1.530 x² = √x →3.03
a) $E_R \approx 3.03$ N/C	
$\beta = \tan^{-1}(1.530 / 2.615)$ x+, y+ → cuadrante I: θ = β	1.530 ÷ 2.615 = 2nd tan⁻¹ →30.32
b) θ ≈ 30.32° con el eje x	

⚠ sen295° es **negativo** (E_2 apunta hacia abajo). Calculadora en grados: 60 cos = 0.5

2do parcial · II · Distancia entre masas

gravitación

m_1 = 18 kg, m_2 = 24 kg, $F = 5\times10^{-12}$ N, $G = 6.67\times10^{-11}$ N·m²/kg². Hallar d.

✍ A MANO	📱 CALCULADORA
$F = G\cdot m_1\cdot m_2 / d^2$ $d = \sqrt{(G\cdot m_1\cdot m_2 / F)}$	
Números: 6.67·18·24 = 2881.44	6.67 × 18 × 24 = →2881.44
Exponentes: $10^{-11} / 10^{-12}$ $= 10^{-11+12} = 10^1$	
$d^2 = (2881.44 / 5)\times10^1$ $= 576.288\times10 = 5762.88$	2881.44 ÷ 5 = →576.288 × 10 = →5762.88
$d = \sqrt{5762.88}$	√x →75.91
d ≈ 75.91 m	